

Will Varner

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Education

University of Georgia

Athens, GA

Bachelor of Science in Computer Science; GPA: 3.8

Expected May 2027

Honors: UGA Presidential Scholar, Zell Miller Scholar

Awards & Leadership: Order of Omega, Roblox Campus Ambassador

Publications

- **W. Varner**, Y. Shao. "Closed-Loop Red-Teaming of Vision-Language-Action Models via Dynamic Co-Simulation." *In preparation for IEEE Intelligent Vehicles Symposium (IV 2027)*.

Experience

Roblox

San Mateo, CA

Software Engineering Intern, Engineering Acceleration

May 2026 – August 2026

- Architected an agentic pipeline to reconstruct declarative service contracts and behavioral specifications from AST source analysis, runtime telemetry, and design docs, replacing 9,700 lines of brittle deterministic parsers.
- Integrated declarative behavior contracts into Roblox's LangGraph and C#/.NET autonomous test-generation platform, lifting critical behavioral obligation coverage by +36.4% in a controlled same-commit, identical-budget A/B benchmark.
- Surfaced 7 owner-confirmed production defects missed in code review, including an 884-day-old inverted boolean cross-serving public endpoint data; drove 3 merged production patches and removed 2,000+ lines of dead routing logic.
- Engineered a differential evaluation harness using order-reversed mutation fixtures to eliminate LLM judge bias and strip framework bloat, benchmarking specification drift across 5 frontier model tiers.
- Drove self-serve platform adoption across 40+ production repositories in 7 engineering orgs via a single-command CLI/Slack workflow, merging 14 service specifications covering 661 behaviors unprompted by external teams.

Mobility Systems Control Laboratory

Athens, GA

Undergraduate AV Research Lead

January 2025 – Present

- Architected a closed-loop red-teaming pipeline coupling a 20Hz C++ telemetry daemon with CARLA 0.9.15 and SUMO co-simulation via POSIX shared-memory IPC, eliminating network socket latency.
- Integrated NVIDIA Alpamayo 2 Super (34B VLA), streaming multi-camera feeds and telemetry to evaluate vision-language reasoning tokens and control trajectories under continuous simulation.
- Engineered an adversarial scenario mutator in Python that dynamically perturbs pedestrian crossing vectors and occlusion angles conditioned on real-time ego vehicle kinematics across 40+ collision topologies.
- Uncovered safety-critical alignment failure modes where the 34B network accurately verbalized imminent collision hazards yet simultaneously output contradictory steering actuation commands.

Leidos

Atlanta, GA

Software Engineering Intern

May 2025 – January 2026

- Ported 15+ backend nodes from CARMA-CARLA Bridge to ROS 2 (rclpy), eliminating legacy bridge middleware to achieve direct, sub-5ms communication with the CARMA Platform.
- Engineered carla_ros2_bridge in C++ and Python, authoring real-time sensor and vehicle interfaces for CARLA 0.10.0 to enable direct, high-throughput ROS 2 co-simulation.
- Prototyped a unified XML-RPC protocol replacing TraCI for actor and sensor synchronization, reducing serialization overhead by 50% across interaction pipelines.

Technical Skills

- **Languages:** C++, C, Python, Go, C#, SQL, Lua / Lua, TypeScript
- **Simulation & Robotics:** ROS 2 (rclpy), CARLA (LibCarla), SUMO, POSIX Shared Memory IPC
- **AI & Backend Infrastructure:** LangGraph, Pydantic, gRPC, .NET, OpenCV
- **Systems & Observability:** Linux/Unix, Docker, VictoriaMetrics, Kibana, CI/CD (GitHub Actions), AWS, GCP